Statistical distributions

Experimental

i.e. Dealing with collected data.

Chp1: Data

Methods of sampling, types of data, and populations vs samples.

Chp2: Measures of Location/Spread

Statistics used to summarise data, including mean, standard deviation, quartiles, percentiles. Use of linear interpolation for estimating medians/quartiles.

Chp3: Representation of Data

Producing and interpreting visual representations of data, including box plots and histograms.

Chp4: Correlation

Measuring how related two variables are, and using linear regression to predict values.

Theoretical

Deal with probabilities and modelling to make inferences about what we 'expect' to see or make predictions, often using this to reason about/contrast with experimentally collected data.

Chp5: Probability

Venn Diagrams, mutually exclusive + independent events, tree diagrams.

Chp6: Statistical Distributions

Common distributions used to easily find probabilities under certain modelling conditions, e.g. binomial distribution.

Chp7: Hypothesis Testing

Determining how likely observed data would have happened 'by chance', and making subsequent deductions.

1: General Probability Distributions

"Given that $P\left(X = x\right) = \frac{k}{x}$, find the value of k."

2: Binomial Distribution

"I toss an unfair coin, with probability heads of 0.6, 10 times. What's the probability I see 5 heads?"

3: Cumulative Binomial Probabilities

"I toss an unfair coin, with probability heads of 0.6, 10 times. What's the probability I see at most 3 heads?"

You are already familiar with the concept of **variable** in statistics: a collection of values (e.g. favourite colour of students in the room):

$\mathbf{x}$	red	green	blue	orange
$\mathbf{P(X = x)}$	0.3	0.4	0.1	0.2

If each is assigned a probability of occurring, it becomes a **random variable**.

✎ A **random variable X represents a single experiment/trial.** It consists of outcomes with a probability for each.

i.e. X is a random variable (capital letter), but x is a particular outcome.

$$P(X = x)$$

"The probability that…" …the outcome of the random variable X… …was the specific outcome x"

A shorthand for $P\left(X = x\right)$ is ✎ $p\left(x\right)$ (note the lowercase p).
It's like saying "the probability that the outcome of my coin throw was heads" ($P(X = heads)$) vs "the probability of heads" ($p(heads)$). In the latter the coin throw was implicit, so we can skip the '$X =$'.

Probability Distributions vs Probability Functions 5

There are two ways to write the mapping from outcomes to probabilities:

The "{" means we have a 'piecewise function'. This just simply means we choose the function from a list depending on the input.

As a function:

$$p(x) = \begin{cases} 0.1x, & x = 1, 2, 3, 4, \\ 0, & \text{otherwise.} \end{cases}$$

e.g. if $x = 2$, then the probability is $0.1 \times 2 = 0.2$

Advantages of functional form:

?

e

As a table:

x	1	2	3	4
p(x)	?			

The table form that you know and love.

Advantages of table form:

?

Example 6

The random variable X represents the **number of heads when three coins are tossed.**

Underlying Sample Space

?

Distribution as a Table

?

Distribution as a Function

?

(Hint: Use your knowledge that $\Sigma p\,(\ldots) = 1$)

Edexcel S1 May 2012

1. A discrete random variable X has the probability function

$$P(X = x) = \begin{cases} k(1-x)^2 & x = -1,\ 0,\ 1 \text{ and } 2 \\ 0 & \text{otherwise.} \end{cases}$$

(*a*) Show that $k = \frac{1}{6}$.

(3)

x	−1	0	1	2
p(x)	?	?	?	?

?

x	2	3	4	5
p(x)	0.1	0.3	0.2	0.4

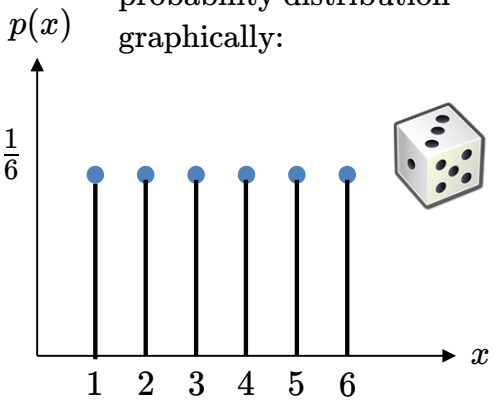
Determine:

$P\,(X > 3) =$?

$P\,(2 \leq X < 4) =$?

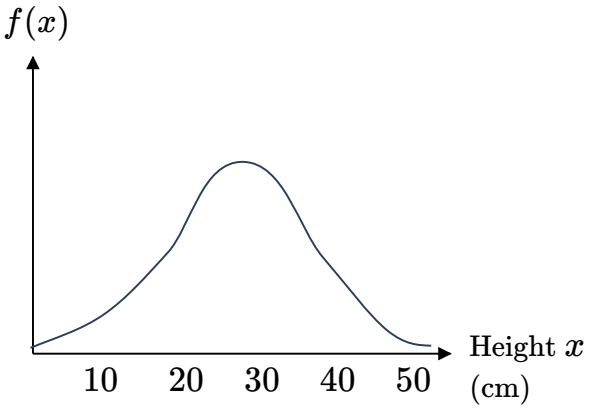
$P\,(2X + 1 \geq 6) =$?

We can also represent a probability distribution graphically:



The throw of a die is an example of a **discrete uniform distribution** because the probability of each outcome is the same.

$p(x)$ for discrete random variables is known as a **probability mass function**, because the probability of each outcome represents an actual 'amount' (i.e. mass) of probability.



We can also have probability distributions for **continuous** variables, e.g. height.

However, the probability that something has a height of say **exactly** 30cm, is infinitely small (effectively 0).

$p(x)$ (written $f(x)$) for continuous random variables is known as a **probability density function**. $p(30)$ wouldn't give us the probability of being 30cm tall, but the amount of probability **per unit height**, i.e. the density. This is similar to histograms where frequency density is the "frequency per unit value". Just as an area in a histogram would then give a frequency, and area under a probability density graph would give a probability (mass).

You will encounter the **Normal Distribution** in Year 2, which is an example of a continuous probability distribution.

5 The random variable X has a probability function

$$P(X = x) = kx \qquad x = 1, 2, 3, 4.$$

Show that $k = \frac{1}{10}$.

?

6 The random variable X has a probability function

$$P(X = x) = \begin{cases} kx & x = 1, 3 \\ k(x-1) & x = 2, 4 \end{cases}$$

where k is a constant.

a Find the value of k.

b Find $P(X > 1)$.

a?

b?

7 The discrete random variable X has a probability function

$$P(X = x) = \begin{cases} 0.1 & x = -2, -1 \\ \beta & x = 0, 1 \\ 0.2 & x = 2 \end{cases}$$

a Find the value of β.

b Construct a table giving the probability distribution of X.

c Find $P(-1 \leqslant X < 2)$.

a?

b?

C?

10 A discrete random variable X has the probability distribution shown in this table.

x	0	1	2	3
$P(X = x)$	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{8}$

Find:

a $P(1 < X \leqslant 3)$ **(1 mark)**

b $P(X < 2)$ **(1 mark)**

c $P(X > 3)$ **(1 mark)**

a?

b?

c?

11 A biased coin is tossed until a head appears or it is tossed four times.

If $P(\text{Head}) = \frac{2}{3}$:

a Write down the probability distribution of S, the number of tosses, in table form. **(4 marks)**

b Find $P(S > 2)$. **(1 mark)**

a?

B?

13 Marie says that a random variable X has a probability distribution defined by the following probability mass function:

$$P(X = x) = \frac{2}{x^2}, \quad x = 2, 3, 4$$

a Explain how you know that Marie's function does not describe a probability distribution. **(2 marks)**

b Given that the correct probability mass function is in the form $P(X = x) = \frac{k}{x^2}, \quad x = 2, 3, 4$ where k is a constant, find the exact value of k. **(2 marks)**

a?

b?

Extension 14

[Challenge p88] The Independent Random Variables X and Y have probability distributions $P(X = x) = \frac{1}{8}, \quad x = 1, 2, 3, 4, 5, 6, 7, 8$ $P(Y = y) = \frac{1}{y}, \quad y = 2, 3, 6$

Find $P(X > y)$.

a?

Let's simplify the problem by using just 3 people:

The probability a randomly chosen person is left-handed is 0.1. If there is a group of 3 people, what is the probability that:

a) All 3 are left-handed.

b) 0 are left-handed.

c) 1 person is left-handed.

d) 2 people are left-handed.

a ?

b ?

c ?

d ?

Let's try to generalise!

If there were x 'lefties' out of 3, then we can see, using the examples, that the probability of a single matching outcome is $0.1^x \times 0.9^{3-x}$.

How many rows did we have each time? In a sequence of three L's and R's, there are "3 choose x", i.e. $\binom{3}{x}$ ways of choosing x of the 3 letters to be L's. Therefore the probability of x out of 3 people being left handed is:

$$\binom{3}{x} 0.1^x 0.9^{3-x}$$

You can model a random variable X with a binomial distribution $B(n, p)$ if

- there are a fixed number of trials, n,
- there are two possible outcomes: 'success' and 'failure',
- there is a fixed probability of success, p
- the trials are independent of each other

If $X \sim B(n, p)$ then:

$$P(X = r) = \binom{n}{r} p^r (1-p)^{n-r}$$

← In our example, 'success' was 'leftie'.

← r is the number of successes out of n.

← "$\sim$" means "has the distribution"

On a table of 8 people, 6 people are left handed.

a) Suggest a suitable model for a random variable X: the number of left-handed people in a group of 8, where the probability of being left-handed is 0.1.

b) Find the probability 6 people are left handed.

c) Suggest why the chosen model may not have been appropriate.

a ? $B(8, 0.1)$

b ?)02268

c ? y.

In general, choosing a well-known model, such as a Binomial distribution, makes certain **simplifying assumptions**. Such assumptions simplifies the maths involved, but potentially at the expense of not adequately modelling the situation.

The random variable $X \sim B\left(12, \frac{1}{6}\right)$. Find:

a) $P(X = 2)$

b) $P(X = 9)$

c) $P(X \leq 1)$

a ?

b ?

c ?

Mental Tip: The two powers add up to n.

Tip: Remember the two 'edge cases':

$P(X = 0) = (1 - p)^n$

$P(X = n) = p^n$

Edexcel S2 June 2010 Q6

A company claims that a quarter of the bolts sent to them are faulty. To test this claim the number of faulty bolts in a random sample of 50 is recorded.

(a) Give two reasons why a binomial distribution may be a suitable model for the number of faulty bolts in the sample. (2)

?

1 $X \sim B(6, 0.2)$

What is $P(X = 2)$?

?

What is $P(X \geq 5)$?

?

2 I have a bag of 2 red and 8 white balls. X represents the number of red balls I chose after 5 selections (with replacement).

a How is X distributed?

?

b Determine the probability that I chose 3 red balls.

?

6 The probability of a switch being faulty is 0.08. A random sample of 10 switches is taken from the production line.

a Define a suitable distribution to model the number of faulty switches in this sample, and justify your choice. **(2 marks)**

b Find the probability that the sample contains 4 faulty switches. **(2 marks)**

a?

b?

7 A particular genetic marker is present in 4% of the population.

a State any assumptions that are required to model the number of people with this genetic marker in a sample of size n as a binomial distribution. **(2 marks)**

b Using this model, find the probability of exactly 6 people having this marker in a sample of size 50. **(2 marks)**

a?

b?

8 A dice is biased so that the probability of it landing on a six is 0.3. Hannah rolls the dice 15 times.

a State any assumptions that are required to model the number of sixes as a binomial distribution. State the distribution. **(2 marks)**

b Find the probability that Hannah rolls exactly 4 sixes. **(2 marks)**

c Find the probability that she rolls two or fewer sixes. **(3 marks)**

a?

b?

c?

Often we wish to find the probability of a range of values.
For a Binomial distribution, this was relatively easy if the range was narrow, e.g. $P(X \leq 1) = P(X = 0) + P(X = 1)$, but would be much more computationally expensive if we wanted say $P(X \leq 6)$.

$$\text{If } X \sim B(10, 0.3), \text{ find } P(X \leq 6).$$

How to calculate

On Classwiz: Press Menu then 'Distributions'.
- Choose "Binomial CD" (the C stands for 'Cumulative'). Choose 'Variable'.
- $x = 6$, $N = 10$, $p = 0.3$

On CG50: Menu Statistics, Dist, Binomial, Bcd
- Set Data=variable, Lower=0, upper =6,
- Numtrial=10, p=0.3

On fx-9860GIII: Statistics, Dist, Binomial, Bcd
- Set Data=variable
- - $x = 6$, $N = 10$, $p = 0.3$

In all calculator types press "=" to see the results

Using tables (e.g. Page 204 of textbook)

Look up $n = 10$ and the column $p = 0.3$.
Then look up the row $x = 6$.
The value should be 0.9894.

Important Note: The tables only have limited values of p. You may have to use your calculator. You will need to use your calculator in the exam anyway.

Cumulative Probabilities

The random variable $X \sim B(20, 0.4)$. Find:

Look up $n = 20, p = 0.4, x = 7$

$P(X \leq 7) =$?

$P(X < 6) =$?

$P(X \geq 15) =$?

Note that the table requires $\leq$

To get this right, just say in your head "*What's the opposite of 'at least 15'?*". Hopefully you can see it's '*at most 14*'.

Given that $X \sim B(25, 0.25)$

$P(X = 6) =$?

$P(X > 20) =$?

$P(6 < X \leq 10) =$?

X can be 7 to 10. So we want up to 10, with everything up to 6 excluded.

Quickfire Questions

Write the following in terms of cumulative probabilities, e.g. $P(X < 7) = P(X \leq 6)$

$P(X < 5) =$?

$P(X \geq 7) =$?

$P(X > 7) =$?

$P(10 \leq X < 20) =$?

$P(10 \leq X \leq 20) =$?

$P(X = 100) =$?

$P(20 < X < 30) =$?

at least 30 = ?

greater than 30 = ?

More Challenging Example

Q

An awkward boyscout boy ventures into a girl guide camp. He asks 20 girls out on the date. The probability that each girl says yes is 0.3.
Determine the probability that he will end up with:
a) Less than 6 girls on his next date.
b) At least 9 girls on his next date.

The boy considers the evening a success if he dated at least 9 girls that evening.
He repeats this process across 5 evenings.
c) Calculate the probability that he had at least 4 successful evenings.
(Note: You won't be able to use your table for (c) as p is not a nice round number – calculate prob directly)

a ?

b ?

c ?

This is an interesting problem because the probability from a Binomial distribution is then used as the p of a **second** separate Binomial distribution.

Q A spinner is designed so that probability it lands on red is 0.3. Jane has 12 spins.
a) **Find the probability that Jane obtains at least 5 reds.**
Jane decides to use this spinner for a class competition. She wants the probability of winning a prize to be < 0.05. Each member of the class will have 12 spins and the number of reds will be recorded.
b) Find how many reds are needed to win the prize.

?

STEP 1: Represent the sentence using probability.

STEP 2: Ensure LHS involves $\leq$ inside probability.

STEP 3: Rearrange.

STEP 4: Use table backwards to find value corresponding to closest probability.

Test Your Understanding 26

Q At Camford University, students have 20 exams at the end of the year. All students pass each individual exam with probability 0.45. Students are only allowed to continue into the next year if they pass some minimum of exams out of the 20.
What do the university administrators set this minimum number such that the probability of continuing to next year is at least 90%?

?

7 The random variable $X \sim B(50, 0.40)$. Find:

a the largest value of k such that $P(X \leqslant k) < 0.05$ **(1 mark)**

b the smallest number r such that $P(X > r) < 0.01$. **(2 marks)**

a?

b?

8 The random variable $X \sim B(40, 0.10)$. Find:

a the largest value of k such that $P(X < k) < 0.02$ **(1 mark)**

b the smallest number r such that $P(X > r) < 0.01$ **(2 marks)**

c $P(k \leqslant X \leqslant r)$. **(2 marks)**

a?

b?

c?

9 In a town, 30% of residents listen to the local radio.
Ten residents are chosen at random.
X = the number of these 10 residents that listen to the local radio.

a Suggest a suitable distribution for X and comment on any necessary assumptions. **(2 marks)**

b Find the probability that at least half of these 10 residents listen to local radio. **(2 marks)**

c Find the smallest value of s so that $P(X \geqslant s) < 0.01$. **(2 marks)**

a?

b?

c?

10 A factory produces a component for the motor trade and 5% of the components are defective. A quality control officer regularly inspects a random sample of 50 components. Find the probability that the next sample contains:

a fewer than 2 defectives **(1 mark)**

b more than 5 defectives. **(2 marks)**

The officer will stop production if the number of defectives in the sample is greater than a certain value d. Given that the officer stops production less than 5% of the time:

c find the smallest value of d. **(2 marks)**

a?

b?

C?